

Investment Cases for Disaster Risk Reduction and Resilience

Enabling Resilience Investment to
enable economic growth and protect
the regions: Future-proofing
infrastructure and communities in the
Bega Valley Shire

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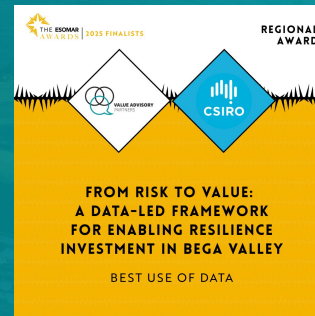


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ESOMAR Awards
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Infrastructure Planning through a Multi-Hazard and Resilience Lens: Accessible Practices to Achieve New Ways of Planning for the Future

ENABLING RESILIENCE INVESTMENT (ERI) FRAMEWORK



ERI Co-Design Process



Resilient Investment
Cases (RIC)



Resilient Investment
Case Explorer (RICE) Tool

Current Infrastructure Planning Practices

- Inconsistent integration of hazard assessments into infrastructure investments
- Reliance on only user benefits for infrastructure business cases
- Limited community and local stakeholder input for resilient infrastructure funding
- Existing planning frameworks reactive, asset-focused, and short-term
- Limited internal process to prioritise resilience and future-proof investments

Connected Engagement through Planning Processes

Accessible, Inclusive,
Usable

Delivering multiple
workshops and working
groups to:

- Convene dialogue spaces and identify stakeholders
- Facilitate adaptive decision making
- Enable active learning

Incorporating Resilience in Infrastructure Planning

Source: Value Advisory Partners 2025

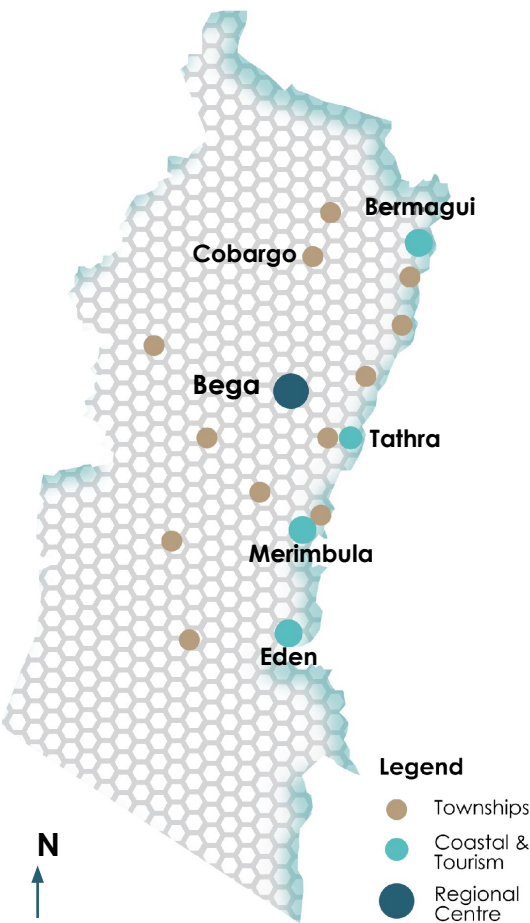
CHALLENGES FOR THE FUTURE OF INFRASTRUCTURE PLANNING FOR TOWNS AND REGIONAL SETTLEMENTS

Challenge for Bega Valley's local communities exposed to natural hazards

Key Challenges

- Bega Valley Shire is an expansive Local Government Area of **over 6,000 kilometres squared** and population of over 36,000 in a **mix of community typologies**
- Impacts from a series of natural hazards that have occurred within the Shire pose a **significant challenge for local communities' livelihoods and well-being**
- Key **regional employment and tourism centres** of Bermagui, Bega, Merimbula, and Eden are especially affected (see map)
- Smaller township communities within the shire also experience **disconnection from services** and can be isolated during hazard events

Communities within Bega Valley Shire



Employment Total
(2024)
17,100

Population Total
(2024)
36,600

Tourism Visitation
(2019)
1,000,000

Source: Value Advisory Partners
2025,
<https://profile.id.com.au/bega-valley/>,
<https://www.tra.gov.au/en/region/local-government-area-profiles>

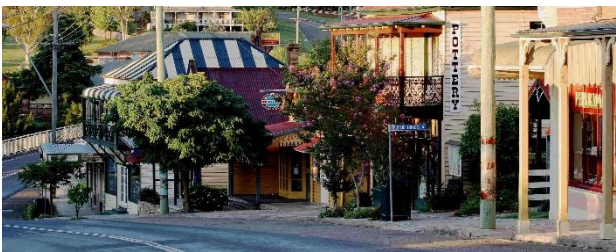
Community Typologies and examples



Regional Centre: Bega



Coastal & Tourism: Merimbula



Townships: Cobargo

Increased Frequency and Intensity of Multiple Hazards having an Overwhelming Effect on Towns and Regional Settlements

Key challenges

- The Bega Valley region has been directly hit with **catastrophic natural disasters within the last 5 years**
- This is through **events from multiple hazards**, with some specific areas within the region hit by multiple events in same year
- Asset damage and ongoing vulnerability from hazards **have negative economic and social impacts across the Shire**
- Infrastructure Planning needs to consider **more frequent and intense hazards** when planning for changes in land use and housing settlement
- Including multi-hazard assessments in place-based infrastructure decisions can lead to more **resilient outcomes** for regions

Examples of recent hazards and asset damage in the Bega Valley Shire

Bushfire: Cobargo



Source: <https://www.theguardian.com/australia-news/2020/jan/05/paradise-lost-locals-flee-eden-as-australia-bushfires-rage-towards-them>

Storm Surge: Tathra



Source: <https://www.begadistrictnews.com.au/story/8498169>

Riverine Flooding: Bega



Source: <https://www.begadistrictnews.com.au/story/7641314/bega-river-flood-warning-downgraded-from-major-to-minor-updated-bega-valley-road-closures/>

Landslides: Brown Mountain

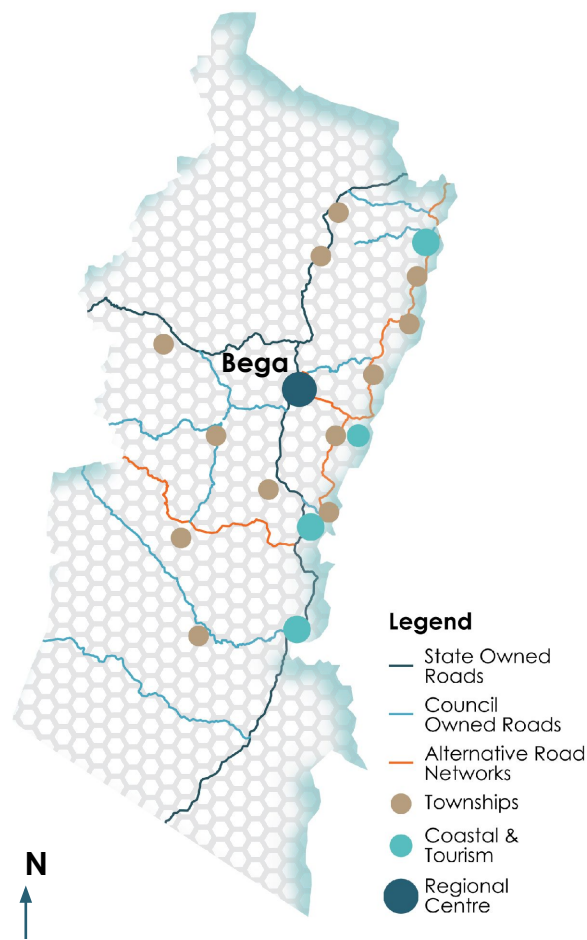


Source: <https://www.begadistrictnews.com.au/story/161343/questions-raised-over-mountain-roads/>

Challenge for Bega Valley's local critical infrastructure and road transport exposed to natural hazards

Need for change to ensure a resilient infrastructure network

- Extensive network of road assets distributed over a large geographic area are **expensive to operate and maintain**, particularly when impacted by recurring hazards
- Some aging, heritage-listed assets limit movement within the Shire, and can be **extremely susceptible to hazards** (e.g. single-lane timber bridges)
- Some coastal communities can become **isolated during hazard events** due to 'one road in, one road out'
- Road infrastructure supports a broad range of service needs, including **access to health and education, as well as tourism and freight**



Source: Value Advisory Partners 2025

Impacts of road network damage on Bega Valley

Connectivity and Liveability

- Evacuation routes for communities cut off
- Access to local services compromised, especially in townships with limited services
- Disconnection to family and friends

Productivity and Industry

- Delays along freight routes
- Industry of specialisation within Bega Valley incl. forestry and fishing being cut off from key economic nodes
- Businesses losing productivity during and after hazard event

Tourism

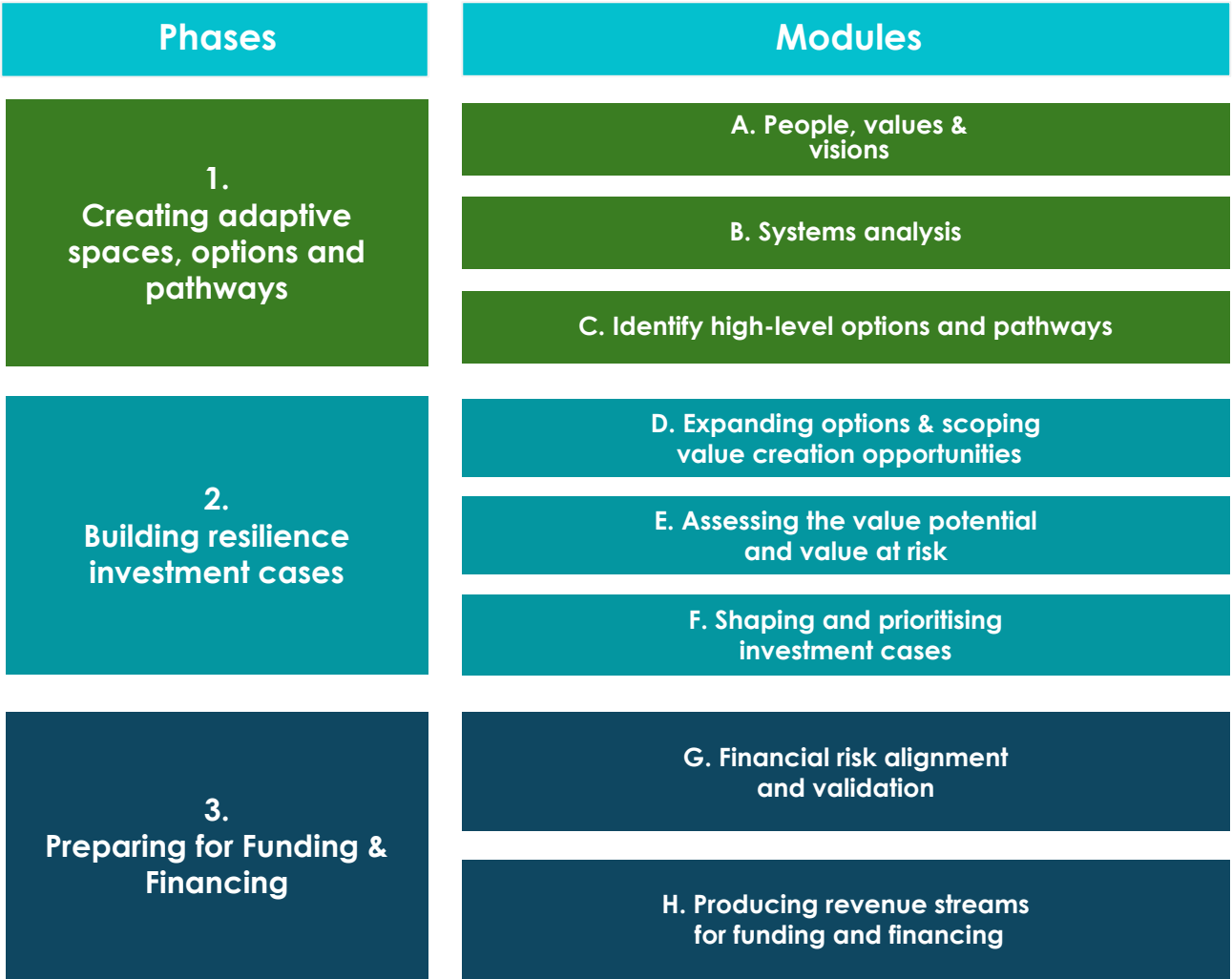
- Key tourism towns and businesses cut off from travellers
- Tourists staying in local towns unable to leave during hazard events e.g Black Summer Bushfires

Enabling Resilience Investment (ERI) Framework

The ERI approach is an engagement, planning and analysis process, underpinned by integrated analytical capabilities and expertise, which incorporates value creation and systemic risk mitigation into the design and delivery of current and future investments.

Benefits of the ERI approach include:

- **Understand climate change risks** and vulnerabilities to places, assets and service delivery, and identify mitigation options
- Evaluate **physical and financial sustainability of critical infrastructure** and assets under a changing climate
- **Enhanced assessment** of damages, losses, co-benefits and costs to prioritise adaptation and mitigation options and build the case for investment
- **Inform future land use, development and infrastructure** plans informed by climate and hazard risks
- **Identify funding sources and pathways** across both public and private sectors arising from resilient investment



Source: Source: Wise et. Al (2022). Enabling Resilience Investment Guidance version 1. CSIRO and Value Advisory Partners

INNOVATION AND TRANSFORMATIVE PROCESSES WITHIN THE ENABLING RESILIENCE INVESTMENT (ERI) PROJECT FOR BVSC



Objectives for ERI within Bega Valley Shire project in future-proofing infrastructure and making communities resilient

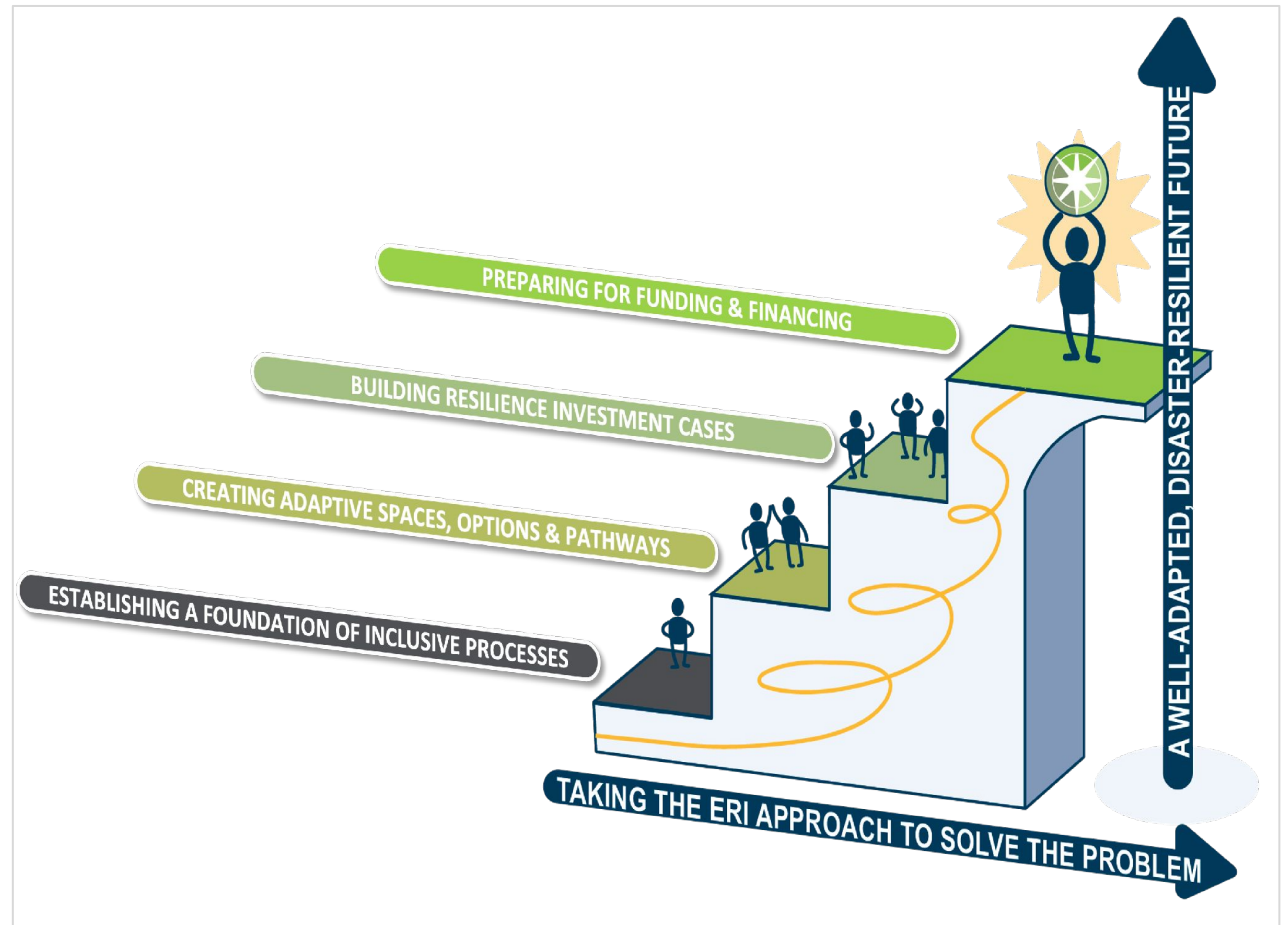
Key Objectives For the Bega Valley Project

- Develop and design **resilience investment cases** that account for a broader range of financial and non-financial benefits, vulnerability and disaster risk reduction and funding and finance options
- Build the **capability and capacity of BVSC and relevant stakeholders**, incorporating the ERI tools and processes so they can lead and engage in **adaptive pathways planning** for disaster risk reduction in Bega Valley Shire
- Create a **broader authorising and enabling environment** for key infrastructure projects to mitigate future extremes to **protect Bega Valley communities and businesses**

The diagram on the right shows how ERI process and was aimed to be embedded in the Bega Valley Shire project that led to incorporating:

- Community values
- Explicitly considered hazard risk
- Ensuring increased probability of success for funding resilient infrastructure

Step Changes to Enable Resilient Investment in Bega Valley Shire



Source: Wise et. Al (2022). Enabling Resilience Investment Guidance version 1. CSIRO and Value Advisory Partners



Development of knowledge sharing and collaboration activities within the Bega Valley Shire project

Key Processes & Outputs

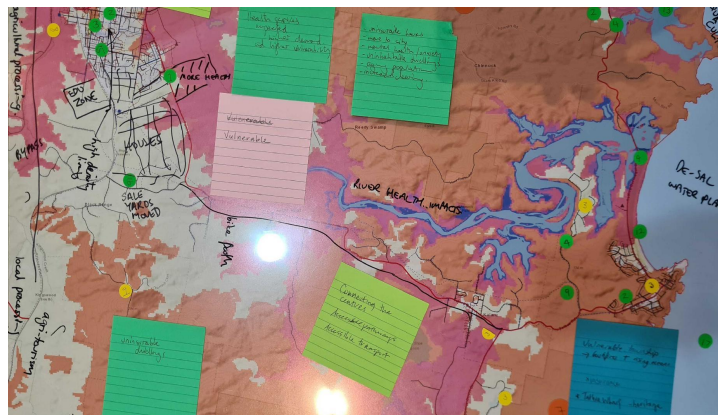
- The ERI approach supports **collaboration and co-design** to generate investible opportunities for a future that delivers values that all stakeholders benefit from
- The engagement process for the project involved **five workshops** with a diverse stakeholder group to:
 - Understand the rationale of the ERI process
 - Frame the vision of a resilient Bega Valley Shire
 - Identify the key service needs
 - Generate resilience-building interventions
 - Stress test interventions under climate change and growth scenarios
- Some examples included within the co-design process are shown on the right building toward a **shared understanding of the vision and objectives** of the project for multiple stakeholders

Examples of knowledge sharing and co-design through BVSC project

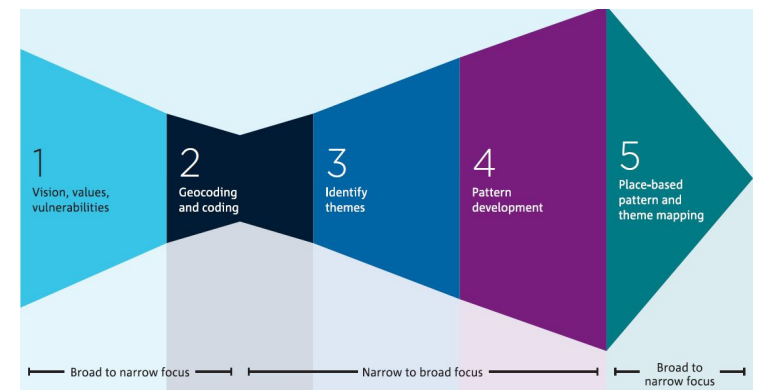
Identifying values & impacts



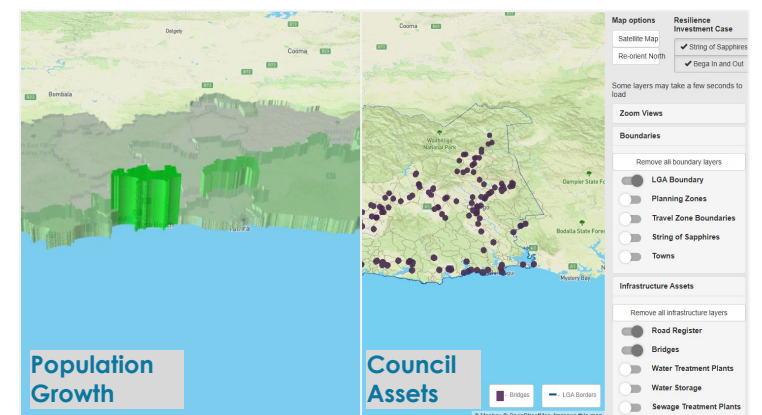
Spatial Mapping using local knowledge



Pattern and themes identification



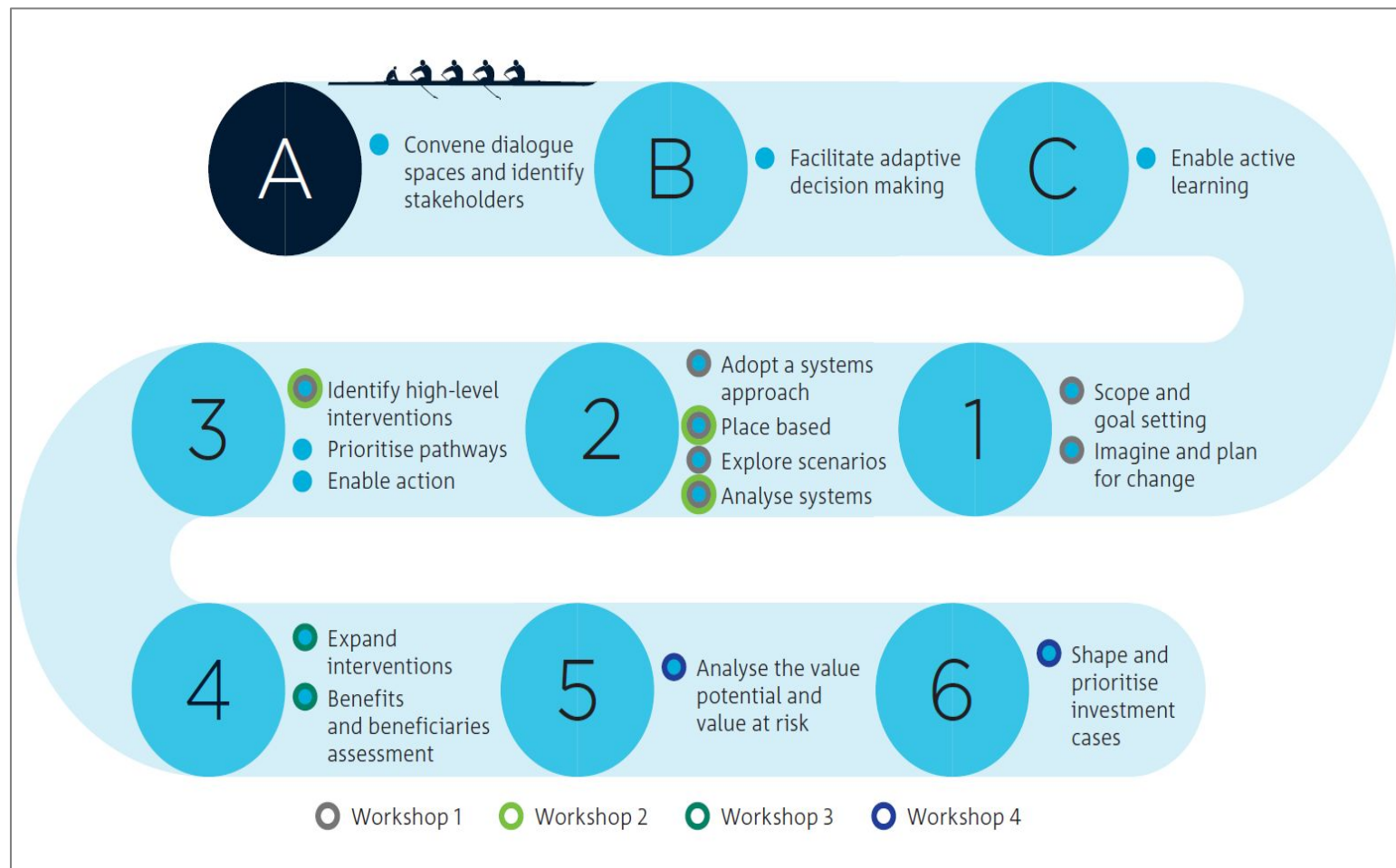
Data Visualisation Tools





Outcomes from the ERI Co-Design Process – Rivermap for navigating stages of a resilience project

- Through the Co-Design process, an innovative outcome was the delivery of a manual for the Council
"Enabling Resilience Investment: A practical manual for Bega Valley Shire Council"
- This included a Rivermap (see summary diagram on right): a clear summary of the ERI process from the project and how it can **apply to future projects**
- This **step-by-step adoption guide** and visual journey supports ongoing ERI implementation and **enhanced organizational capability**
- This allowed the project to **not just be a one-off example of the process** but rather a **step-change in decision making** for long-term resilience investment



Source: Meharg et. Al (2024). Enabling Resilience Investment: a practical manual for Bega Valley Shire Council.



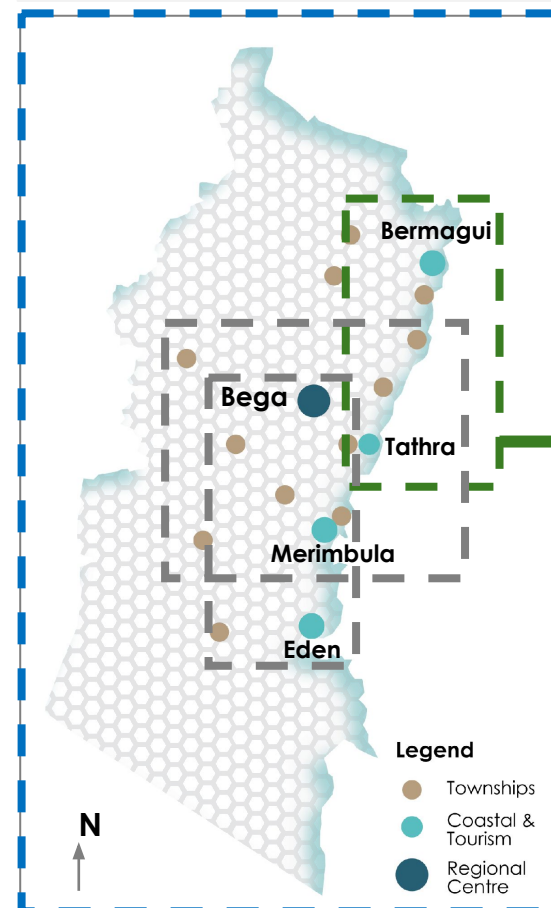
Development of Resilience Investment Cases (RIC) within the ERI Framework

- A Resilience Investment Case is a description of a **resilience investment logic and packaged set of resilience-building interventions** aimed at addressing clearly defined place-based service needs.
- It does not target a specific funding source but documents the **logic that underpins a potential investment decision**, before specific solutions are identified a decision is made.
- The logic and package of interventions can then be **targeted to specific funding sources** as opportunities arise.
- A Multi-criteria assessment of four candidate focal areas, each with its own specific service needs, resulted in **two prioritised focal areas to be developed into RICs** (highlighted on the right)

Attributes of a Resilience Investment Case

- Have a **system, place-based, and values-focused** problem framing
- Focus on **assets, services & beneficiaries in place**
- Generate **multiple options and adaptation pathways**
- Develop climate-change compatible objectives delivered through **value protection and value creation**
- Develop investment cases to enact bundles of options that **broaden funding sources and deliver resilience** benefits over time

Two Resilient Investment Cases Selected for BVSC



RIC #1

Resilient Transport Corridors

- Regional focus across LGA
- Ensuring continued road access for goods, service and people
- Resilience for going in and out of Bega Valley during hazards and stability.

RIC #2

String of Sapphires

- Focus on connectivity between coastal villages between Bermagui and Tathra
- Investment as a resilient alternate route to current main route of Princes Highway
- Importance for both local and tourist connectivity and liveability

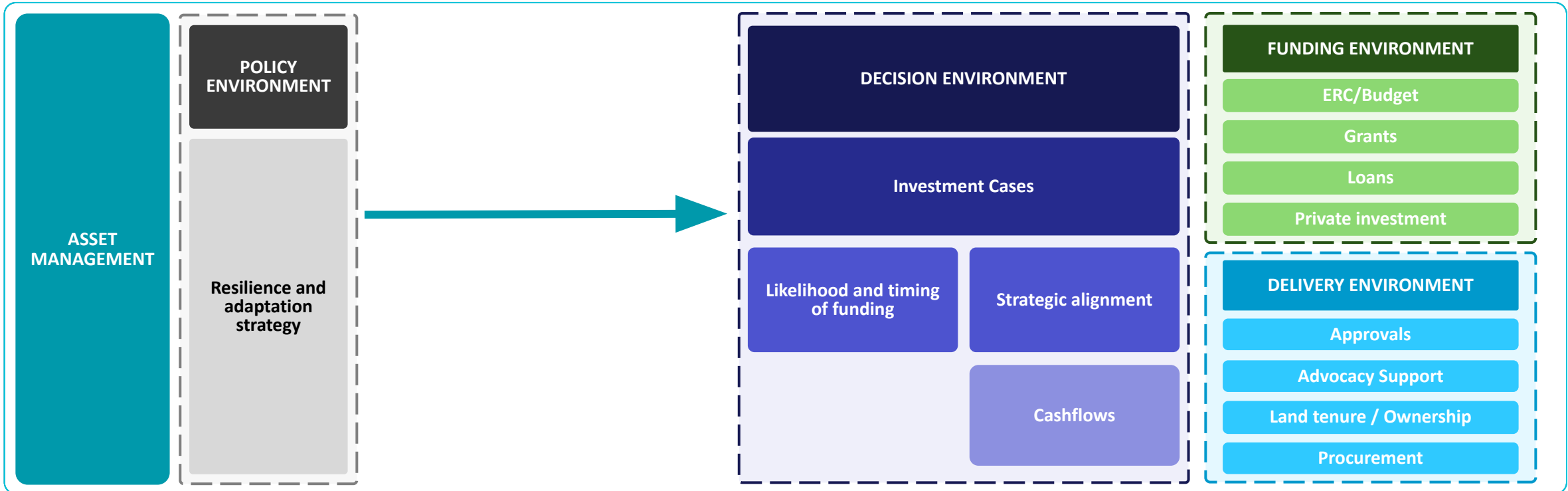
Source: Value Advisory Partners 2025; Xenarios et. Al (2024). Enabling Resilience Investment in the Bega Valley – Final Report



Application of Resilient Investment Cases (RIC) of infrastructure projects within Bega Valley Shire

- Traditional approaches to infrastructure investment are often single asset end-of-life focussed, with economic assessment for direct user benefits, assuming risks are known and quantifiable (i.e. static climate)
- A Resilience Investment Case differs in that it considers whole-of-life for multiple assets, and takes a place-based systems and services view
- The benefits considered are more diverse, include resilience and adaptation outcomes, and use scenarios to factor in uncertainty
- The RICE tool puts the key information in one platform to allow decision makers to test assumptions and create their own bundles for funding

Current Process: Infrastructure Business Case

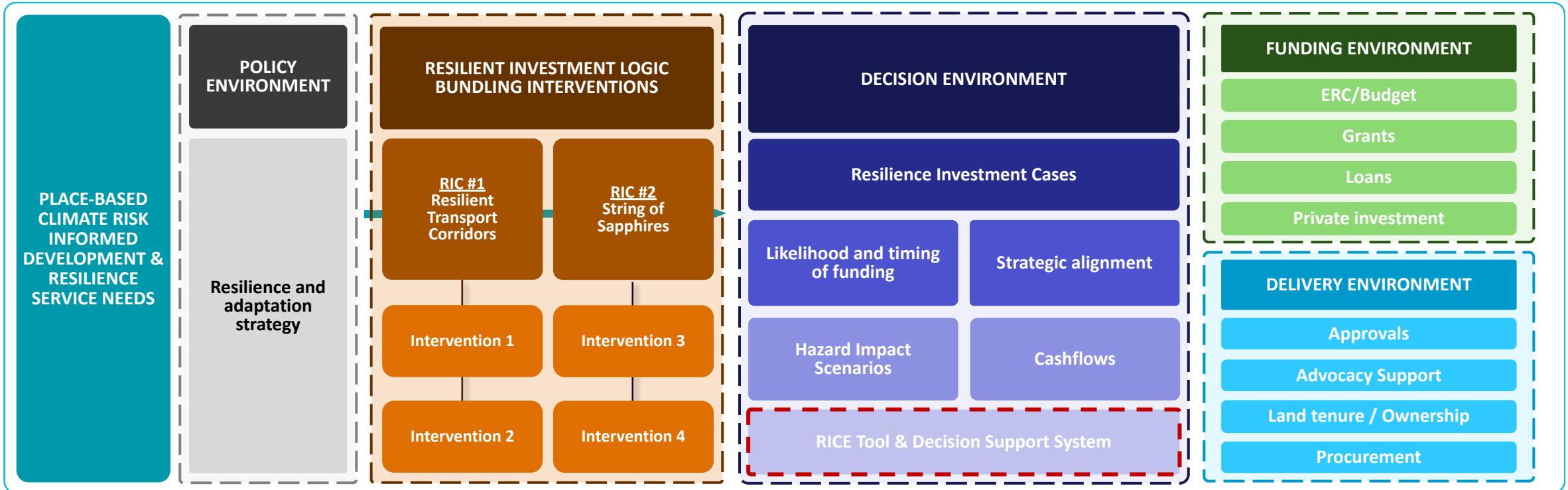




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Process used in Bega Valley Project: Resilient Investment Case (RIC)





Development of a Dynamic Decision Support System to be used as a part of the RIC Process

- A key output of the Bega Valley Shire ERI project was a **web-based interactive decision support tool** – the Resilience Investment Case Explorer (RICE)
- The RICE tool incorporates all interventions identified through the project **quantifying benefits, costs, hazard susceptibility, and reconstruction value** of current assets (value at risk)
- The diagram on the right shows **the multiple capabilities and features** the RICE tool includes.
- This meant there was **a common platform for decision making** allowing dynamic **options assessment in real time**
- The next two slides will show two examples of how the RICE tool can be used:
 - **Detailed assessment:** ability to test the investment logic and create bundles of interventions and opportunities
 - **Summary of evidence:** One page summary showing key outcomes for inclusion in funding applications and advocacy

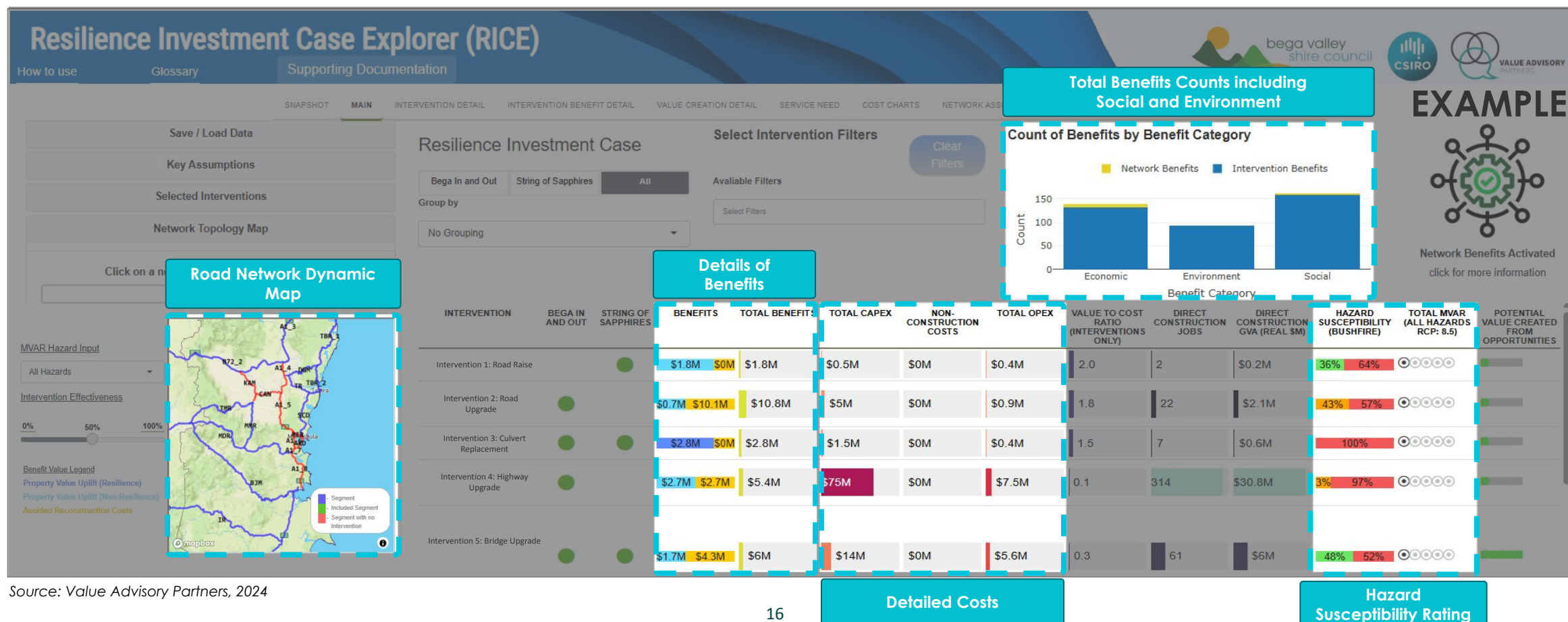


Source: Adapted from Resilience Investment Cases – Systems model, Meharg et. Al (2024). Enabling Resilience Investment: a practical manual for Bega Valley Shire Council.



Application of RICE Tool to facilitate place-based decision-making through detailed assessment

- RICE provides a view on **how interventions spatially interrelate**, and of the potential **broader network benefits** that arise from the **combination of specific interventions**
- It allows the ability to **test the investment logic** and create **bundles of interventions and opportunities** for potential funding, including benefit value, avoided damages, susceptibility, location and type of intervention, benefits and beneficiaries, and service need descriptions.



Source: Value Advisory Partners, 2024

Outcomes from the ERI Project for Bega Valley Shire Council

Key Challenges	Transformative approaches for ERI	Project Outcomes
- Existing planning frameworks reactive, asset-focused, short-term, and don't adequately address climate risk - Input from broader stakeholders (internal and external) is constrained and/or siloed	ERI Co-Design Process 	Participatory workshops with 30+ council staff and external stakeholders built shared understanding of values, risk, and resilience objectives ERI Manual & Rivermap: step-by-step adoption guide and visual journey supports ongoing ERI implementation and enhanced organizational capability
- Increased frequency and intensity of multiple hazards exposing systemic vulnerabilities - Limited internal processes to prioritise resilience and future-proof infrastructure investments	Resilient Investment Cases (RIC) 	- Development of two case studies: Resilient Transport Corridors, and String of Sapphires Coastal Connectivity Clearer rationale for resilience intervention actions reflecting short- and long-term hazard risks addressing specific service needs
- Constrained funding, growing community expectations - Need for more objective and evidence-based way to evaluate costs and benefits, and prioritise asset investment	Resilient Investment Case Explorer (RICE) Tool 	Common framework for decision making allowing dynamic options assessment in real time Facilitating bundling and sequencing of interventions into adaptable investment pathways and targeted funding proposals

Reflection from BVSC project for Resilient Investment Cases

Lessons Learnt

- Complex change processes, such as ERI, require ongoing executive **leadership, engagement, commitment and continuity to deliver**
- The project was an **effective pilot process which identified resilience investment** interventions for funding by combining many process steps, tools and methods together with underlying competencies.
- Bundling of interventions/investment options based on **place and service need can strengthen funding applications.**
- Wide range of engagement methods and approaches is needed to sustain engagement and co-design throughout the project.
it is important to **establish working groups early in the project.**



Further opportunities from BVSC project for Resilient Investment Cases

Opportunities and Next Steps

Incorporate enhanced processes from ERI project into other local council infrastructure projects

The transformative processes shown in the road focused infrastructure project could be applied in other projects run by the council such as other types of infrastructure, and knowledge sharing could occur within existing local government partnerships

Potential Opportunities: Departments within Bega Valley Shire Council, Canberra Region Join Organisation (CJRO)

Transforming Processes Within all levels of government

Expanding transformative processes within BVSC project to other parts of government to increase capability and knowledge in planning for ongoing and uncertain changes in natural hazards, and establishing ongoing cross-sector governance for strategic assessments

Potential opportunities : Commonwealth and State Government Departments for Employment and Industry, Land Use Management, Transport and Economic Development

Inclusion within Disaster Adaptation Plans

Processes within BVSC project could be included within Disaster Adaptation Plans connecting strategic planning policies to resilience and adaptation planning, to improve better infrastructure in well located and protected areas

Potential opportunities : Ongoing strategies for NSW Reconstruction Authority, Emergency Management Victoria

A New Model for Strategic planning in Urban and Regional Areas

BVSC as an innovative project shows how to implement similar transformative planning processes by conducting risk-informed, integrated strategic assessments in the preparation of long-term planning into other areas within Australia

Potential Opportunities: Six Cities Regions Plans (NSW), Greater Adelaide Regional Plan



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